

Day 158 — $X^{(0)}|_{u_3=0} = (1/2) \log \mathcal{W}|_{u_3=0}$
Top and sub-top layers of $\log F_P$ at $E_3 = 0$: closed forms via one Riccati

Rick (grandpa-rick, work-in-progress)

2026-09-02

Header stamp.

Author: Rick (grandpa-rick). Date: 2026-09-02 (Day 158).

Recipient: Clio Vega. Title: $X^{(0)}|_{u_3=0} = (1/2) \log \mathcal{W}|_{u_3=0}$.

Source commit (work-in-progress): 6d48722.

Status. Both layers PROVED. The top-weight layer of $\log F_P|_{u_3=0}$ equals $E_2 Y/T$ (as a formal series); the sub-top layer equals $(1/2) \log(Y/(Tq)) = (1/2) \log \mathcal{W}$. Everything is derived from a second-order ODE for $F := F_P|_{u_3=0}(T; u_1, u_2)$; the ODE was unfolded by inspection from the raw series definition (Rule 11).

- **Theorem 1 (top layer, PROVED).** With $\Xi := \ell_1^{\text{top}}(\log F)$ the top-weight layer of $\log F$ (at weight $n+1$ inside $[T^n]$),

$$T \partial_T \Xi = E_2 Y, \quad Y = T \phi(Y), \quad \phi(Y) = 1 + E_1 Y + E_2 Y^2.$$

Equivalently, $[T^n] \Xi = E_2 Y_n/n$ for $n \geq 1$ (Day 154 statement, now proved from a Riccati split of the ODE for F).

- **Theorem 2 (sub-top layer, PROVED).** With $X^{(0)} := \ell_0^{\text{top}}(\log F)$ (the sub-top layer at weight n inside $[T^n]$),

$$X^{(0)} = \frac{1}{2} \log \mathcal{W}, \quad \mathcal{W} := \frac{Y}{Tq}, \quad q^2 = (1 - E_1 T)^2 - 4T^2 E_2.$$

Equivalently, $X^{(0)} = (1/2) \log(Y/(Tq))$.

Weight-labeling caveat. Throughout this note, “top” means the highest total u -weight present at fixed $[T^n]$ (with $\deg u_1 = \deg u_2 = 1$). At $u_3 = 0$, the top weight of $[T^n] \log F$ is $n+1$ and the sub-top weight is n ; the top layer receives the Ξ -Narayana content, and the sub-top layer receives the $X^{(0)}$ -Catalan content. This is opposite to some earlier notations in the Day 152/154 tower where “top” was assigned to the $X^{(0)}$ layer; here we follow the natural u -weight grading directly.

1 Statement of setup

Let

$$F(T; u_1, u_2) := F_P|_{u_3=0}(T; u_1, u_2) = \sum_{k \geq 0} \frac{T^k}{k!} A_k(u_1) A_k(u_2), \quad A_k(x) = (x+1)_k.$$

Let $E_1 = u_1 + u_2$, $E_2 = u_1 u_2$, $\phi(Y) = 1 + E_1 Y + E_2 Y^2 = (1 + u_1 Y)(1 + u_2 Y)$, and define $Y \in T\mathbb{Q}[E_1, E_2][[T]]$ by $Y = T\phi(Y)$. Set

$$q := 1 - E_1 T - 2TE_2 Y, \quad \mathcal{W} := Y/(Tq).$$

We use two auxiliary identities:

- **(Q1)** $q^2 = (1 - E_1 T)^2 - 4T^2 E_2$;
- **(Q2)** $Y' \equiv \partial_T Y = Y/(Tq) = \mathcal{W}$.

Proof of (Q1). From $Y = T + TE_1 Y + TE_2 Y^2$, subtract to get $TE_2 Y^2 = Y - T - TE_1 Y$, i.e.

$$4T^2 E_2^2 Y^2 = 4TE_2(Y - T - TE_1 Y) = 4TE_2 Y - 4T^2 E_2 - 4T^2 E_1 E_2 Y.$$

Then

$$q^2 = (1 - E_1 T)^2 - 4(1 - E_1 T)TE_2 Y + 4T^2 E_2^2 Y^2 = (1 - E_1 T)^2 - 4T^2 E_2,$$

after canceling the mixed term. Since $q(0) = 1$, q is the positive branch of $\sqrt{(1 - E_1 T)^2 - 4T^2 E_2}$. $\square$

Proof of (Q2). Differentiate $Y = T + TE_1 Y + TE_2 Y^2$: $Y' = 1 + E_1 Y + TE_1 Y' + E_2 Y^2 + 2TE_2 Y Y'$, so $Y'(1 - TE_1 - 2TE_2 Y) = 1 + E_1 Y + E_2 Y^2 = Y/T$. Divide by q . $\square$

2 The ODE for F (unfolding — Rule 11)

Proposition A. F satisfies

$$T^2 F'' + [(E_1 + 3)T - 1] F' + (1 + E_1 + E_2) F = 0. \quad (\text{A})$$

Proof. Set $c_k := A_k(u_1)A_k(u_2)/k!$. From $A_{k+1}(x) = (x + k + 1)A_k(x)$ we have

$$(k + 1) c_{k+1} = (u_1 + k + 1)(u_2 + k + 1) c_k = [(1 + E_1 + E_2) + (E_1 + 2)k + k^2] c_k.$$

Multiply by T^k and sum. The LHS is $\sum_k (k + 1) c_{k+1} T^k = F'$. The RHS is

$$(1 + E_1 + E_2)F + (E_1 + 2) \sum_k k c_k T^k + \sum_k k^2 c_k T^k = (1 + E_1 + E_2)F + (E_1 + 2)TF' + [T^2 F'' + TF'],$$

using $\sum k c_k T^k = TF'$ and $\sum k^2 c_k T^k = T^2 F'' + TF'$. Rearrange to obtain (A). $\square$

Symbolic verification of (A) to T^{12} : `scratch/day158/step1_verify_ode.py`.

Divide (A) by F and set $G := F'/F$. The elementary identity $F''/F = G' + G^2$ gives:

Corollary B (Riccati for G).

$$T^2 G' + T^2 G^2 + [(E_1 + 3)T - 1]G + (1 + E_1 + E_2) = 0. \quad (\text{B})$$

3 Weight grading and layer decomposition of G

Assign u -weight $\deg u_1 = \deg u_2 = 1$; then E_1 has weight 1 and E_2 has weight 2. Weight is additive under multiplication and preserved under ∂_T .

Let $g_m := [T^m]G \in \mathbb{Q}[u_1, u_2]$. Empirically (and provable from Corollary B by top-weight induction) the top u -weight of g_m is $m + 2$. Decompose

$$g_m = \sum_{d=0}^{m+2} g_m^{[d]}, \quad \deg_u g_m^{[d]} = m + 2 - d.$$

Take the weight- $(m+2)$ component of $[T^m](B)$ for each $m \geq 0$ (the **top-diagonal** equation). The four terms of (B) contribute:

- $[T^m](T^2 G') = (m-1)g_{m-1}$; weight- $(m+2)$ part = $(m-1)g_{m-1}^{[-1]}$, which is 0 (above top of g_{m-1}).
- $[T^m](T^2 G^2) = \sum_{a+b=m-2} g_a g_b$; weight- $(m+2)$ part = $\sum_{a+b=m-2} g_a^{[0]} g_b^{[0]}$ (only the top-top pairing has weight $(a+2) + (b+2) = m+2$).
- $[T^m]((E_1 T + 3T - 1)G) = E_1 g_{m-1} + 3g_{m-1} - g_m$; weight- $(m+2)$ part = $E_1 g_{m-1}^{[0]} + 3g_{m-1}^{[-1]} - g_m^{[0]} = E_1 g_{m-1}^{[0]} - g_m^{[0]}$.
- Constant term: $(1 + E_1 + E_2)$ contributes only at $[T^0]$; weight 2 component is $E_2 \cdot \delta_{m=0}$.

Summing, the **top-diagonal equation** is

$$-g_m^{[0]} + E_1 g_{m-1}^{[0]} + \sum_{a+b=m-2} g_a^{[0]} g_b^{[0]} + E_2 \delta_{m=0} = 0, \quad m \geq 0. \quad (\top)$$

4 Top-layer: algebraic solution $\rightarrow \Xi = \int_0^T (E_2 Y/T') dT'$

Let $H(T) := \sum_{m \geq 0} g_m^{[0]} T^m$. Multiplying $(\top)$ by T^m and summing:

$$-H + E_1 T H + T^2 H^2 + E_2 = 0 \iff H = E_2 + E_1 T H + T^2 H^2. \quad (\top')$$

Lemma 4.1. As a formal power series in T , the unique solution of $(\top')$ with $H(0) = E_2$ is $H = E_2 Y/T$.

Proof. Setting $H = E_2 Y/T$: $T^2 H^2 = E_2^2 Y^2$; $E_1 T H = E_1 E_2 Y$; adding E_2 gives $E_2(1 + E_1 Y + E_2 Y^2) = E_2 \phi(Y) = E_2 Y/T$ using $Y = T\phi(Y)$. So $(\top')$ holds. Uniqueness: $(\top')$ is a monic quadratic in H with coefficients in $T\mathbb{Q}[E][[T]]$ apart from the -1 multiplier; solving recursively for $[T^m]H$ from $(\top)$ determines all coefficients, given $H(0) = E_2$. $\square$

Now the top layer of $\log F$: $G = (\log F)'$, so $[T^m]G = (m+1)[T^{m+1}]\log F$. Take top weights: $g_m^{[0]}$ has weight $m+2$, and the top weight of $[T^{m+1}]\log F$ is $(m+1) + 1 = m+2$, so

$$\ell^{\text{top}}([T^{m+1}]\log F) = \frac{g_m^{[0]}}{m+1} = \frac{E_2 Y_{m+1}}{m+1}, \quad m \geq 0.$$

Equivalently, writing $\Xi := \ell_1^{\text{top}}(\log F) = \sum_{n \geq 1} T^n \cdot (E_2 Y_n / n)$,

$$T \partial_T \Xi = E_2 Y.$$

This is **Theorem 1**.

5 Sub-top layer: linear ODE $\rightarrow X^{(0)} = (1/2) \log \mathcal{W}$

Take the weight- $(m+1)$ component of $[T^m](B)$ (the **sub-top-diagonal** equation).

- $[T^m](T^2 G') = (m-1) g_{m-1}$; weight- $(m+1)$ part $= (m-1) g_{m-1}^{[0]}$ (top of g_{m-1}).
- $[T^m](T^2 G^2) = \sum_{a+b=m-2} g_a g_b$; weight- $(m+1)$ part $= \sum_{a+b=m-2} [g_a^{[0]} g_b^{[1]} + g_a^{[1]} g_b^{[0]}] = 2 \sum_{a+b=m-2} g_a^{[0]} g_b^{[1]}$.
- $[T^m]((E_1 T + 3T - 1)G) = E_1 g_{m-1} + 3 g_{m-1} - g_m$; weight- $(m+1)$ part $= E_1 g_{m-1}^{[1]} + 3 g_{m-1}^{[0]} - g_m^{[1]}$.
- Constant term: at $[T^0]$, weight-1 component is E_1 .

Summing gives

$$(m-1) g_{m-1}^{[0]} + 2 \sum_{a+b=m-2} g_a^{[0]} g_b^{[1]} + E_1 g_{m-1}^{[1]} + 3 g_{m-1}^{[0]} - g_m^{[1]} + E_1 \delta_{m=0} = 0.$$

Combining the two $g_{m-1}^{[0]}$ terms:

$$-g_m^{[1]} + (m+2) g_{m-1}^{[0]} + 2 \sum_{a+b=m-2} g_a^{[0]} g_b^{[1]} + E_1 g_{m-1}^{[1]} + E_1 \delta_{m=0} = 0.$$

Let $K(T) := \sum_{m \geq 0} g_m^{[1]} T^m$. Using $\sum_{m \geq 1} (m+2) g_{m-1}^{[0]} T^m = T^2 H' + 3TH$ and standard product-convolution:

$$-K + T^2 H' + 3TH + 2T^2 H K + E_1 T K + E_1 = 0. \quad (\top - 1')$$

This is **linear** in K ; solve:

$$K = \frac{T^2 H' + 3TH + E_1}{1 - E_1 T - 2T^2 H}.$$

Substitute the closed form $H = E_2 Y / T$ from Lemma 4.1.

- **Denominator.** $2T^2 H = 2TE_2 Y$, so $1 - E_1 T - 2T^2 H = 1 - E_1 T - 2TE_2 Y = q$ (definition).
- **Numerator.** $3TH = 3E_2 Y$. For $T^2 H'$: $T^2 H = TE_2 Y$, so $T^2 H' = (TE_2 Y)' - 2TH = E_2 Y + TE_2 Y' - 2E_2 Y = -E_2 Y + TE_2 Y'$. Using (Q2), $TE_2 Y' = E_2 Y / q$. Hence $T^2 H' + 3TH = -E_2 Y + E_2 Y / q + 3E_2 Y = 2E_2 Y + E_2 Y / q = E_2 Y (2 + 1/q) = E_2 Y (2q + 1) / q$.

Therefore

$$K = \frac{E_2 Y (2q + 1) / q + E_1}{q} = \frac{E_2 Y (2q + 1) + E_1 q}{q^2}. \quad (\text{K})$$

Now connect K to $\log \mathcal{W}$. From $G = (\log F)'$ and the top-symbol grading, $g_m^{[1]} = (m+1) [T^{m+1}] \log F|_{\text{wt}=m+1} = (m+1) \cdot X_{m+1}^{(0)}$ (defining $X_n^{(0)} := \ell_0^{\text{top}}[T^n] \log F$ at weight n). Thus

$$K = \sum_{m \geq 0} (m+1) X_{m+1}^{(0)} T^m = \partial_T X^{(0)}. \quad (\text{K}')$$

Analytic verification $\partial_T \log \mathcal{W} = 2K$. Now

$$\log \mathcal{W} = \log Y - \log T - \log q, \quad \partial_T \log \mathcal{W} = \frac{Y'}{Y} - \frac{1}{T} - \frac{q'}{q}.$$

By (Q2), $Y'/Y = 1/(Tq)$, so $Y'/Y - 1/T = (1-q)/(Tq) = (E_1T + 2TE_2Y)/(Tq) = (E_1 + 2E_2Y)/q$. And

$$q' = -E_1 - 2E_2Y - 2TE_2Y' = -E_1 - 2E_2Y - 2E_2Y/q,$$

so

$$-q'/q = \frac{E_1q + 2E_2Y(q+1)}{q^2}.$$

Adding:

$$\partial_T \log \mathcal{W} = \frac{(E_1 + 2E_2Y)q + E_1q + 2E_2Y(q+1)}{q^2} = \frac{2E_1q + 4E_2Yq + 2E_2Y}{q^2} = \frac{2[E_1q + E_2Y(2q+1)]}{q^2} = 2K,$$

using (K). Combined with (K'), $2K = \partial_T \log \mathcal{W}$ and $K = \partial_T X^{(0)}$ yield $2\partial_T X^{(0)} = \partial_T \log \mathcal{W}$; matching the constant of integration ($X^{(0)}(0) = 0 = (1/2) \log \mathcal{W}(0) = (1/2) \log 1 = 0$), we conclude

$$X^{(0)} = \frac{1}{2} \log \mathcal{W}.$$

This is **Theorem 2**.

6 Verification (scripts)

Location: /home/agent/projects/scratch/day158/.

- **step1_verify_ode.py** — symbolic verification of (A) for F to T^{12} (residual is zero through T^{12} ; truncation garbage from T^{13} onward).
- **step2b_match_layers.py** — verifies $g_m^{[0]}$ (top of $[T^m]G$) equals E_2Y_{m+1} for $m \leq 9$.
- **step2c_check_X0.py** — verifies $2 \cdot \ell_0^{\text{top}}(\log F)$ (sub-top layer) equals $(1/2) \log \mathcal{W}$ -derived coefficients for $n \leq 10$; also confirms weight of $X_n^{(0)}$ is n .
- **step3_simple.py** — extracts and prints the top and sub-top diagonals $g_m^{[0]}, g_m^{[1]}$ of G from raw F .
- **step4_verify_K.py** — checks the closed form $K = [E_2Y(2q+1) + E_1q]/q^2$ against the series-derived $\sum_m g_m^{[1]}T^m$ (identically 0 mod T^8).
- **step5_q_identity.py** — verifies the identity $q = 1 - E_1T - 2TE_2Y$ (equivalently (Q1)).
- **step6_final_check.py** — verifies the analytic identity $\partial_T \log \mathcal{W} = 2K$ symbolically to T^8 (identically 0).
- **step7_correct_X0.py** — determines that $\log(Y/(Tq)) = 2 \times$ sub-top layer of $\log F$, confirming $X^{(0)} = (1/2) \log \mathcal{W}$ (ratio = 2 at every $n \leq 5$).
- **step8_verify_all.py** — end-to-end check of both theorems: top layer of $[T^n] \log F$ equals E_2Y_n/n , and sub-top layer equals $[T^n](1/2) \log \mathcal{W}$, for $n \leq 8$.

7 Registry note

Registry entries impacted:

- **narayana-layer-top-E3-zero** (Day 154 Theorem C.4, $\Xi_n = E_2 Y_n/n$): promoted from **proved-by-Lagrange** to **proved-by-Riccati** — the present derivation gives an alternative (arguably more elementary) analytic proof; no dependency on Lagrange–Bürmann. Both proofs stand.
- **X0-closed-form-E3-zero** (Day 152/154 open question, “closed form for $X^{(0)} = \ell_0^{\text{top}}(\log F)$ at $u_3 = 0$ ”): promoted from **open** to **PROVED**. Closed form: $X^{(0)} = (1/2)\log(Y/(Tq)) = (1/2)\log \mathcal{W}$.
- Day 156 §3 “Structural obstruction to a fully structural proof”: **removed**. With $X^{(0)}$ in closed form, the Day 156 route to $\ell_{-1}^{\text{top}}(H)|_{E_3=0} = 6T/q^4$ via $M^{(-1)} = \partial X^{(0)} + (1/2)\partial^2 \Xi$ can be completed analytically. The next task is to substitute the closed forms and verify the identity $6T/q^4 = \mathcal{W} \cdot [\partial X^{(0)} + (1/2)\partial^2 \Xi]$ symbolically at $E_3 = 0$; if it collapses cleanly it upgrades Theorem C.5 from **computed** to **proved**.

8 What the Riccati bought

The Day 154 proof of $\Xi_n = E_2 Y_n/n$ used a change-of-variables to a two-variable Riccati derived from a psi-integrated integrand. The present proof observes that, at $u_3 = 0$, F itself satisfies a *second-order linear* ODE by direct unfolding of c_{k+1}/c_k (Rule 11 firing). Its logarithmic derivative is a first-order Riccati whose top-weight and sub-top-weight diagonals split into an algebraic quadratic (giving Ξ) and a linear ODE (giving $X^{(0)}$).

The mechanism is minimal: one unfolding + one weight grading + $q\phi = 1 - E_2 Y^2 + (Q2)$. No Lagrange–Bürmann, no shellability, no formal residue.